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February 5, 2025

Emily Freilich, M.D., Director
Division of Neurology I
Center for Drug Evaluation and Research
Food and Drug Administration
Attn: Document Control Room
5901-B Ammendale Road
Beltsville, MD 20705-1266

Re: IND No. 167326 Serial No. 0003, PrP-lowering divalent siRNA 2439-exNA

Initial Investigational New Drug Application

Dear Dr. Freilich:

In accordance with 21 CFR 312.23, enclosed is an Investigational New Drug Application for PrP-lowering divalent siRNA 2349-exNA for the indication of prion disease. This investigational drug is an intrathecal oligonucleotide drug designed to bind and degrade the RNA encoding prion protein, which is the root cause of all cases of prion disease.

This original IND application is to request permission to conduct a first-in-human (FIH) clinical trial in symptomatic patients entitled "An open-label, single ascending dose study to evaluate the safety, tolerability, pharmacokinetics and pharmacodynamics of intrathecally administered PrP-siRNA in adult patients with symptomatic prion disease." The primary objective is to evaluate the safety and tolerability of the investigational drug following a single dose administration in patients with active prion disease. Secondary objectives include determining pharmacologic target engagement based on a pharmacodynamic (PD) measurement of CSF PrP concentration.

In addition, we also request permission to conduct a trial in pre-symptomatic individuals at genetic risk, entitled "An open-label, single ascending dose study to evaluate the safety, tolerability, pharmacokinetics and pharmacodynamics of intrathecally administered PrP-siRNA in healthy adult patients at genetic risk for prion disease." Prion disease is typically very rapidly progressive after onset, and even a potent PrP-lowering drug will not be able to reverse neurodegeneration that has already occurred. Our best opportunity to preserve full cognitive function and quality of life is to intervene prior to disease onset, and indeed, our studies in mouse models indicate that there is an outsize benefit to early treatment. My family is personally

affected by genetic prion disease and it is my life's mission to develop a preventive treatment. I believe that pre-symptomatic individuals at risk for prion disease represent a serious imminent disease population and that the risks of the investigational drug are acceptably small relative to the risk of disease onset absent treatment.

All nonclinical, quality (chemistry, manufacturing, controls) and clinical information for 2439-exNA is provided in this IND submission. In accordance with 21 CFR §312.40, clinical investigations will not be initiated until 30 days after the FDA's receipt of this IND.

The confidentiality of this submission, and all information contained herein, is claimed by Broad Institute under all applicable laws and regulations. Disclosure of any such information is not authorized without the prior written authorization of Broad Institute.

This IND submission includes, investigator documentation (1572 and CV) for the single US site, and www.clinicaltrials.gov information via Form 3674.

Any questions concerning this submission can be addressed to me at Broad Institute, 415 Main St. Cambridge, MA 02142 or alternatively via email at [redacted]

Sincerely,

[redacted] Digitally signed by Eric Minikel Date: 2025.02.05 09:52:48 -05'00'

Eric Vallabh Minikel, PhD

Director, Prion Therapeutic Science

Broad Institute of MIT and Harvard

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Cambridge, MA 02142

Electronic Submission Specifications

This submission is being submitted via the Research IND Application Builder within the CDER NextGen Portal. We are not utilizing eCTD submission.